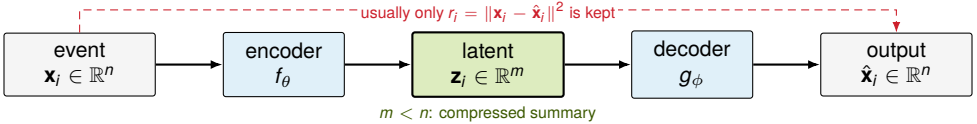
Autoencoder latent space



What the autoencoder learns

$$\mathbf{z}_i = f_\theta(\mathbf{x}_i) \in \mathbb{R}^m, \qquad \hat{\mathbf{x}}_i = g_\phi(\mathbf{z}_i) \in \mathbb{R}^n$$

$$\min_{\theta,\phi} \frac{1}{N} \sum_{i=1}^{N} \left\| \mathbf{x}_i - g_\phi(f_\theta(\mathbf{x}_i)) \right\|_2^2$$

Why look at $\mathbf{z}_i$?

- r_i: one number per event
- $\mathbf{z}_i$: a structured, m-dim description
- Similar r_i can hide very different events

Key message: $\mathbf{z}_i$ carries more information than the 1D error r_i.

Visualizing jet embeddings

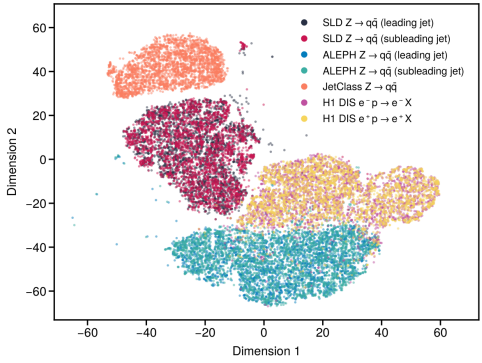


Fig. 6, arXiv:2606.00224: t-SNE visualization of OmniLearned jet embeddings.

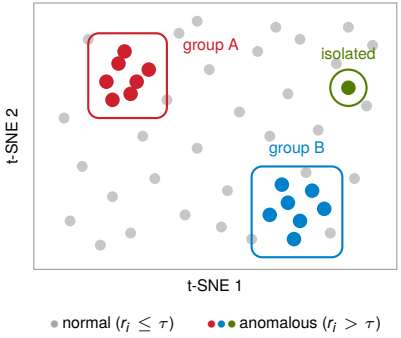
t-SNE of jet embeddings

- Learned jet embeddings are projected to 2D for visualization.
- Jets with similar learned features appear nearby.
- Similar physics topologies group together in embedding space.

Are anomalies alike?

1. Flag events with large error: $r_i > \tau$.
2. Take their latent vectors $\mathbf{z}_i$.
3. Project with t-SNE and highlight the flagged events.

– **Packed together**: anomalies look alike $\Rightarrow$ likely a common source.
– **Isolated**: a one-off unusual event.
– Different clusters $\Rightarrow$ different kinds of anomaly.



Conclusion: r_i flags anomalies; latent space shows which belong together.